

Customer Shopping Behavior Analysis

An End-to-End Retail Analytics Study

Python (EDA) | SQL Analysis | Power BI Dashboard | Business Recommendations

Dataset: customer_shopping_behavior.csv · 3,900 transactions · 18 attributes

1. Project Overview

This project delivers a complete analysis of a retail customer shopping behavior dataset, combining four complementary workstreams: Python-based data cleaning and exploratory data analysis (EDA), SQL-based business question answering, an interactive Power BI dashboard, and data-driven business recommendations. The goal is to understand who the customers are, what and how they buy, which factors drive revenue, and where the business should focus to grow retention and spend.

Objectives

- Profile the customer base by demographics (age, gender, location) and purchasing habits.
- Identify revenue drivers across product categories, items, and customer segments.
- Quantify the impact of discounts, promo codes, subscriptions, and shipping choices on spend.
- Segment customers into New, Returning, and Loyal tiers based on purchase history.
- Translate findings into an interactive Power BI dashboard for ongoing monitoring.
- Produce concrete, prioritized business recommendations.

Tools & Workflow

Tool	Role in this project
Python (pandas, matplotlib)	Data cleaning, feature engineering, exploratory data analysis
PostgreSQL / SQL	Business-question querying on the cleaned dataset (customer_database.sql)
Power BI	Interactive dashboard for KPIs and self-service filtering (dashboard.pbix)
Jupyter Notebook	pandas_file.ipynb — cleaning pipeline and PostgreSQL export

2. Dataset Overview

The raw dataset (**customer_shopping_behavior.csv**) contains 3,900 individual purchase transactions across 18 attributes, covering customer demographics, product details, pricing, shipping, discounts, ratings and purchase history.



Column Reference

Field	Description
Customer ID	Unique transaction/customer identifier
Age	Customer age (years)
Gender	Male / Female
Item Purchased	Specific product name (25 unique items)
Category	Clothing, Footwear, Accessories, Outerwear
Purchase Amount (USD)	Transaction value
Location	U.S. state (50 unique)
Size / Color / Season	Product attributes
Review Rating	Customer rating, 1–5 scale (37 missing, imputed)
Subscription Status	Whether the customer is a subscriber
Shipping Type	Standard, Express, Free Shipping, Next Day Air, etc.
Discount Applied / Promo Code Used	Identical fields — one was dropped as redundant
Previous Purchases	Count of prior purchases by the customer
Payment Method	Credit Card, PayPal, Cash, Venmo, etc.
Frequency of Purchases	Stated purchase cadence (Weekly, Monthly, etc.)

Data Cleaning & Feature Engineering (Python)

- **Missing values:** 37 nulls in Review Rating were imputed using the median rating within each product Category.
- **Standardized column names:** lower-cased and underscored (e.g. 'Purchase Amount (USD)' → purchase_amount).

- **age_group:** engineered via quartile binning (pd.qcut) into Young Adult, Adult, Middle-aged, Senior.
- **purchase_frequency_days:** mapped stated frequency labels (Weekly, Monthly, ...) to a numeric day count.
- **Redundancy check:** confirmed Discount Applied and Promo Code Used were identical for every row, so Promo Code Used was dropped.
- **Export:** the cleaned dataframe was loaded into a PostgreSQL table (public.customer) via SQLAlchemy for the SQL analysis stage.

3. Exploratory Data Analysis (Python)

Key distributions and relationships in the cleaned dataset, visualized with pandas and matplotlib.

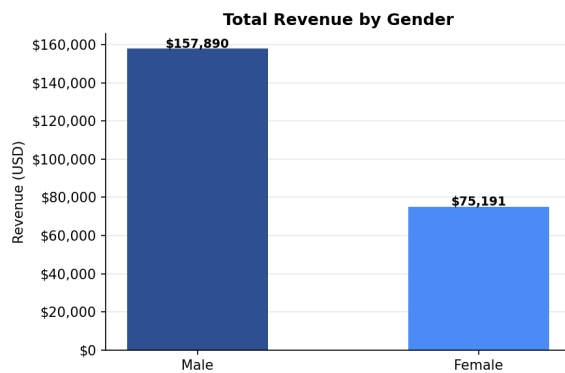


Fig 1. Male customers generate ~2x more revenue than female customers.

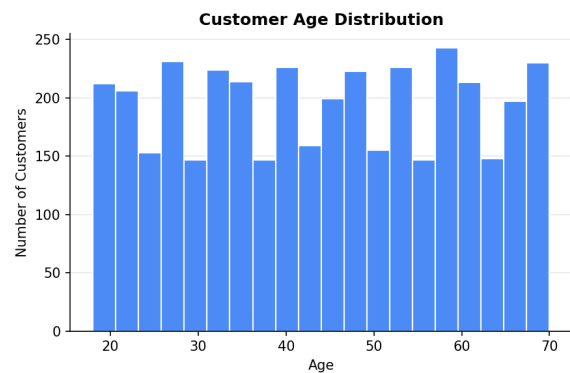


Fig 2. Customer ages are broadly spread from the teens through the 60s.

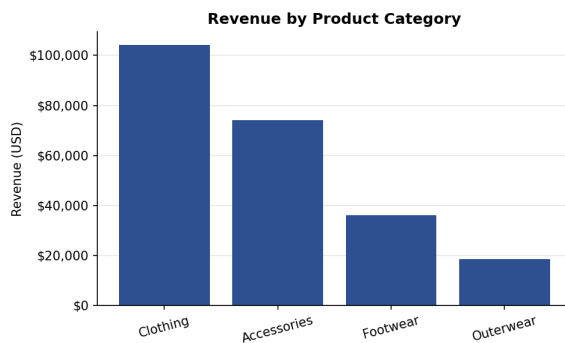


Fig 3. Clothing is the leading revenue category.

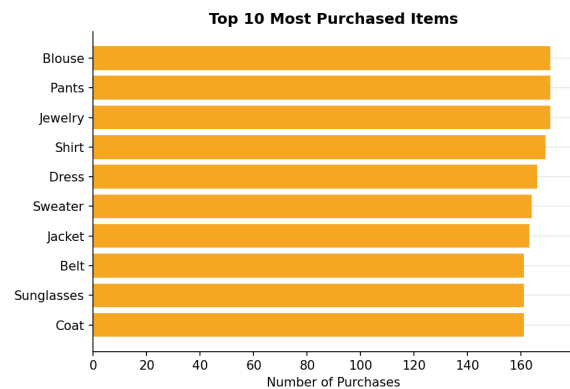


Fig 4. Blouses and jewelry lead purchase volume.



Fig 5. Review ratings cluster around 3.0–4.5 (avg 3.75).

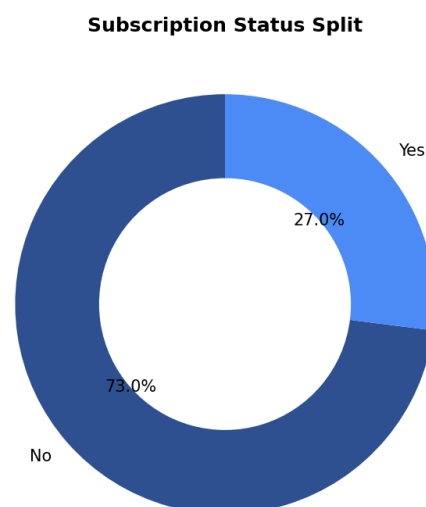


Fig 6. Only 27% of customers are active subscribers.

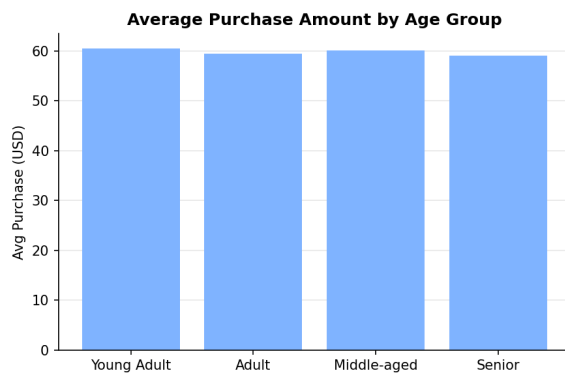


Fig 7. Average spend is stable (~\$59-60) across all age groups.



Fig 9. Discounted purchases average slightly higher spend.

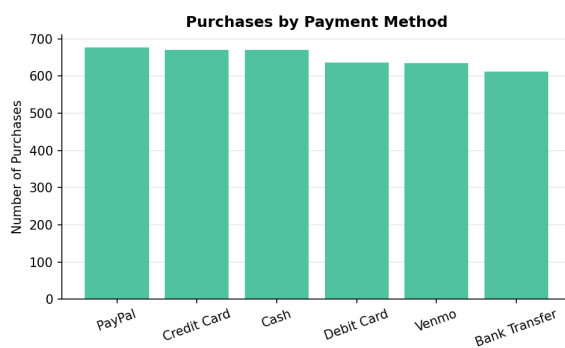


Fig 8. Payment methods are used fairly evenly.

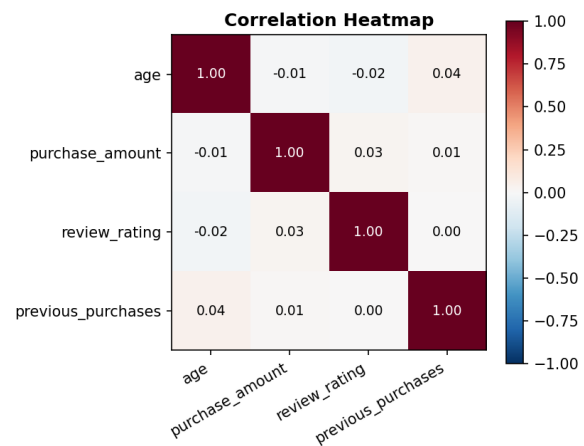


Fig 10. Purchase amount, age, review rating and previous purchases are essentially uncorrelated.

EDA Highlights

- Average purchase amount is \$59.76 (median \$60), indicating a tight, symmetric spend distribution with no extreme skew.
- Male customers account for \$157,890 (68%) of total revenue vs. \$75,191 (32%) for female customers.
- Clothing is the dominant category (\$104K+ in revenue), followed by Accessories, Footwear, and Outerwear.
- Purchase amount shows essentially no linear correlation with age, review rating, or previous purchases — spend level is driven more by product/category choice than by customer profile.
- Only 27% of customers are subscribed, yet non-subscribers already generate the majority of revenue — subscription is under-penetrated relative to its potential value.

4. Data Analysis Using SQL

The cleaned dataset was loaded into PostgreSQL and queried directly (see `customer_database.sql`) to answer ten core business questions. Results below were computed on the identical cleaned data.

Q1. Revenue by Gender

Gender	Total Revenue (USD)
Male	\$157,890
Female	\$75,191

Q2. Discount Users Spending Above Average

839 customers used a discount and still spent at or above the overall average purchase amount of \$59.76 — discounting is not simply attracting low-spend, price-sensitive buyers.

Q3. Top 5 Products by Average Review Rating

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Handbag	3.78

Q4. Standard vs. Express Shipping — Avg Purchase

Shipping Type	Avg Purchase (USD)
Express	\$60.48
Standard	\$58.46

Q5. Subscribers vs. Non-Subscribers

Status	Customers	Avg Spend	Total Revenue
No (non-subscriber)	2,847	\$59.87	\$170,436
Yes (subscriber)	1,053	\$59.49	\$62,645

Subscribers spend almost the same per transaction as non-subscribers — the subscription program is not currently a meaningful lever for higher average order value, only for retention frequency.

Q6. Top 5 Products by Discount Rate

Item	% of Purchases Discounted
Hat	50.0%

Item	% of Purchases Discounted
Sneakers	49.7%
Coat	49.1%
Sweater	48.2%
Pants	47.4%

Q7. Customer Segmentation (New / Returning / Loyal)

Segment	Definition	Customers
New	1 previous purchase	83
Returning	2–10 previous purchases	701
Loyal	11+ previous purchases	3,116

An overwhelming 80% of the customer base already qualifies as "Loyal" by purchase history — retention economics matter more here than new-customer acquisition.

Q8. Top 3 Products per Category

Category	Rank	Item	Orders
Accessories	1	Jewelry	171
Accessories	2	Belt	161
Accessories	3	Sunglasses	161
Clothing	1	Blouse	171
Clothing	2	Pants	171
Clothing	3	Shirt	169
Footwear	1	Sandals	160
Footwear	2	Shoes	150
Footwear	3	Sneakers	145
Outerwear	1	Jacket	163
Outerwear	2	Coat	161

Q9. Repeat Buyers (5+ Previous Purchases) — Subscription Split

Subscription Status	Repeat Buyers
No	2,518
Yes	958

Most repeat buyers are still non-subscribers — high previous-purchase counts do not translate automatically into subscription sign-ups, pointing to a conversion opportunity.

Q10. Revenue by Age Group

Age Group	Total Revenue (USD)
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

Revenue is remarkably even across age groups (each ~24-27% of total) — the customer base is broad rather than concentrated in one age segment.

5. Power BI Dashboard

A companion Power BI report (**dashboard.pbix**) connects to the same cleaned PostgreSQL table (`public.customer`) and turns the analysis into a single interactive page titled **"Customer Behavior Dashboard."** Open the `.pbix` file in Power BI Desktop (with a live connection to the customer table) to explore it interactively.

Dashboard Layout

Visual	What it Shows
KPI Cards (x3)	Average Purchase Amount, Average Review Rating, Number of Customers
Donut Chart	Customer count split by Subscription Status
Column Chart	Total Revenue by Category
Column Chart	Customer Count by Category
Clustered Bar Chart	Total Revenue by Age Group
Clustered Bar Chart	Customer Count by Age Group
Slicers (x4)	Subscription Status, Gender, Category, Shipping Type — for cross-filtering all visuals

How to Use It

- Use the four slicers to filter the whole page — e.g. isolate 'Female' + 'Subscribed' customers to see their category mix and revenue instantly.
- The KPI cards give an at-a-glance health check (avg order value, avg satisfaction, customer count) that updates live as filters are applied.
- The category and age-group visuals are paired (revenue vs. customer count) so volume and value can be read side by side — useful for spotting categories with high traffic but low spend, or vice versa.

Note: this PDF cannot render the live .pbix file directly — Power BI reports are interactive applications, not static documents. Open dashboard.pbix in Power BI Desktop (or Power BI Service) to view and interact with it. The layout and fields above were extracted directly from the report's definition file.

6. Business Recommendations

1. Grow subscription penetration deliberately

Only 27% of customers subscribe, and subscribers don't yet out-spend non-subscribers per order. Bundle subscription with a tangible per-order benefit (early access, free express shipping) rather than a passive badge, and target the 2,518 non-subscribed repeat buyers (Q9) first — they already show loyalty behavior.

2. Protect margin on high-discount items

Hat, Sneakers, Coat, Sweater and Pants are discounted on close to half of all purchases (Q6). Test smaller, more targeted discounts or replace blanket discounting with loyalty-tier pricing, since discount users are not simply price-sensitive — 839 of them already spend above average (Q2).

3. Double down on Clothing, cross-sell into Outerwear

Clothing drives the largest share of revenue and the top items (Blouse, Pants, Shirt) sell in near-equal volume (Q8). Outerwear is the smallest category — bundle it with top Clothing items at checkout to lift average order value.

4. Leverage Express shipping as an upsell

Express shipping customers already spend \$2 more per order on average than Standard (Q4). Test express shipping as a default upsell at checkout, or a threshold-based free-express perk.

5. Treat the base as retention-first, not acquisition-first

80% of customers are already 'Loyal' (11+ previous purchases, Q7). Prioritize retention programs — service quality, loyalty rewards, personalized offers — over new-customer discounting, since the biggest revenue lever is keeping existing high-frequency buyers active.

6. Use ratings to guide merchandising, not to gate it

Top-rated items (Gloves, Sandals, Boots — Q3) are accessory/footwear items rather than the bestselling Clothing lines. Feature these high-satisfaction items more prominently in marketing to reinforce trust, while monitoring lower-rated high-volume items for quality issues.

7. Don't over-index on age-based targeting

Revenue and average spend are nearly flat across all four age groups (Q10, Fig 7). Marketing spend is likely better allocated by behavior (subscription status, purchase frequency, category affinity) than by age bracket alone.

7. Conclusion

This analysis shows a broad, stable customer base (\$233K in total revenue across 3,900 transactions) whose spending is remarkably consistent across age groups and only weakly related to review satisfaction or purchase history. The clearest opportunities are structural rather than demographic: convert the large pool of loyal, non-subscribed repeat buyers into subscribers; rationalize heavy discounting on a handful of items; and use shipping-tier and category bundling to lift average order value. The Power BI dashboard operationalizes these same metrics for ongoing, self-service monitoring by the business team.

Files in this Project

File	Purpose
customer_shopping_behavior.csv	Raw source data (3,900 rows x 18 columns)
pandas_file.ipynb	Python cleaning, feature engineering, PostgreSQL export
customer_database.sql	10 business-question SQL queries
dashboard.pbix	Interactive Power BI dashboard
Customer_Shopping_Behavior_Analysis.pdf	This report